Scope before Reuse: corrected theory sections (draft for review)

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Abstract

Corrected problem formulation, theory and related-work paragraph for FSE-2027-13; see docs/paper/THEORY.md for the derivation and docs/paper/latex/CHANGES.md for what each part replaces.

1 Problem Formulation

Skills, scopes and cells. A skill is $S = \langle \pi, \mathcal{D}, \mathcal{I}, \mathcal{P}, \mathcal{K} \rangle$: a procedure π (a parameterised edit program), dependency and environment fingerprints $\mathcal{D}$, input assumptions $\mathcal{I}$ (executable predicates on the target), observed postconditions $\mathcal{P}$ with the contexts they were observed in, and an interaction footprint $\mathcal{K}$, the code regions and behavioural properties that π reads or writes. A context x is a (repository, revision, environment, task) tuple. A scope family $\mathbb{S} = \{s_1, \ldots, s_L\}$ is a finite partition of contexts declared before any trial. We call a pair $c = (S, s)$ a *cell*; for a set of skills whose footprints form a connected overlap graph, their composition, applied together in scope s, is a cell of its own (for two skills, $(S \circ S', s)$).

Loss and selective transfer risk. For a task x, $\ell(x, \pi) \in [0, 1]$ is its functional loss with the skill applied and $\ell(x, \varnothing)$ the loss without it. We use one minus the held-out pass rate plus 0.5 per regression, *clipped at* 1, so that the range assumption below holds. The paired difference is $Z = \ell(x, \pi) - \ell(x, \varnothing) \in [-1, 1]$.

Definition 1.1 (Selective transfer risk). For a cell $c = (S, s)$ and a distribution D over the contexts of scope s, $\mathrm{STR}_D(c) = \mathbb{E}_{x \sim D}\big[\ell(x, \pi) - \ell(x, \varnothing)\big]$. With P_s the distribution trials are drawn from and Q_s the deployment distribution, a cell is *admissible* if $\mathrm{STR}_P(c) \le -\theta$ for a benefit margin $\theta > 0$, and *harmful* if $\mathrm{STR}_Q(c) > 0$. After a retirement (Section 2) the environment may have changed; epoch k of a cell then has its own $P_s^{(k)}, Q_s^{(k)}$, and every statement below is per epoch.

The definition is an excess over the no-skill baseline, so a skill that neither helps nor harms is not admissible, and it is relative to a declared scope; Proposition 2.4 shows why it cannot be otherwise.

Assumptions.

Assumption 1 (Paired trials). *For every cell c there is an infinite sequence of tasks $x_1^c, x_2^c, \ldots$ drawn i.i.d. from P_s (in practice, drawn* with replacement *from the target's task set; without replacement the conditional mean of the next draw drifts and Lemma 2.1 needs a finite-population e-process instead); each task is run with and without the skill on the same snapshot and seed. Different cells may depend on each other.*

Assumption 2 (Bounded shift). *Deployment contexts of scope s follow Q_s with $\mathrm{TV}(P_s, Q_s) \le \xi_s$, where $\mathrm{TV}(P, Q) = \sup_A |P(A) - Q(A)|$.*

Execution rule. Skills that are executed together form a set E. Split E into the connected components of its footprint-overlap graph (two skills are adjacent when their footprints intersect). We execute E only if every component with two or more skills is itself an admitted cell; a skill whose composite with an already admitted skill is still undecided is not executed together with it. This closes the gap between per-cell certificates and what actually runs.

Admission problem. Given a library, a validation budget and a pre-allocated error budget α, decide which cells may execute, such that $\Pr[\exists \text{ an admitted cell } c : \mathrm{STR}_Q(c) > 0] \le \alpha$ over the whole library, over adaptively chosen cells and over re-admissions after retirement, and such that, under the execution rule, no executed set is harmful except on that event. Theorem 2.2 and Corollary 2.6 give this whenever $2\xi_s \le \theta$ for every scope; for coarser scopes they give the weaker, explicit statement $\mathrm{STR}_Q(c) \le -\theta + 2\xi_s$.

2 Theory

We claim no new statistics: the guarantee is assembled from betting e-processes [9], Ville's inequality [6] and a union bound. What is new is the estimand, the per-cell ledger that covers compositions and re-admission, the proof that the shift term is unavoidable, and the exact characterisation of composition risk.

2.1 Per-cell evidence

Lemma 2.1 (Betting e-process). *Fix a cell c and let $Y_i = -Z_i$. Under $H_0^c : \mathbb{E}[Y] \le \theta$ (the cell is not admissible), the process $K_0 = 1$, $K_n = \prod_{i \le n}\big(1 + \lambda_i(Y_i - \theta)\big)$ with predictable $\lambda_i \in [0, 1/(1+\theta))$ satisfies $\Pr[\exists n : K_n \ge 1/\beta] \le \beta$ for every $\beta \in (0, 1)$.*

Proof. Since $Y_i \ge -1$, every factor is at least $1 - \lambda_i(1+\theta) > 0$. By Assumption 1 and predictability, $\mathbb{E}[K_i \mid \mathcal{F}_{i-1}] = K_{i-1}\big(1 + \lambda_i(\mathbb{E}[Y_i] - \theta)\big) \le K_{i-1}$, so (K_n) is a nonnegative supermartingale with $K_0 = 1$, and Ville's inequality gives the claim. □

The bound holds at any data-dependent stopping time, including the end of the validation budget, which is why a fixed-sample interval would not do.

2.2 Library-wide admission guarantee

Theorem 2.2 (Admission over a declared scope family). *Let C be the countable set of all cells that could ever be formed, compositions included, and let a ledger $\alpha_{c,k} \ge 0$ be fixed in advance, as a function of the cell's identity, for every cell c and epoch $k \ge 0$ (epoch $k \ge 1$ is the k-th re-admission after retirement), with $\sum_{c \in C} \sum_{k \ge 0} \alpha_{c,k} \le \alpha$; for instance $\alpha_{c,k} = \alpha_c \cdot 6/(\pi^2 (k+1)^2)$ with $\sum_c \alpha_c \le \alpha$. A library that grows is covered as long as the share of every future cell is fixed by the same rule. Suppose each epoch uses fresh tasks and admits c only when its e-process (Lemma 2.1) reaches $1/\alpha_{c,k}$, while the choice of cells, their order and the stopping times may depend on all data. Under Assumptions 1 and 2,*

$$\Pr\big[\exists \text{ admitted } (c,k) : \mathrm{STR}_{Q^{(k)}}(c) > -\theta + 2\xi_{s(c)}\big] \le \alpha,$$

where $P^{(k)}, Q^{(k)}$ are the trial and deployment distributions of epoch k. In particular, if $2\xi_s \le \theta$ for every scope, the probability of admitting any harmful cell is at most α.

Proof. For every (c, k) let $A_{c,k} = \{\sup_n K_n^{c,k} \ge 1/\alpha_{c,k}\}$, defined on the whole infinite task sequence of that epoch. An admission

of (c,k) implies $A_{c,k}$; adaptive selection and stopping can only forgo admissions. Hence the event that some cell with H_0^c true is admitted is contained in $\bigcup_{(c,k):H_0^c} A_{c,k}$, whose probability is at most $\sum \alpha_{c,k} \le \alpha$ by Lemma 2.1 and a union bound, which needs no independence across cells. Finally, by Lemma 2.3, $\mathrm{STR}_{Q^{(k)}}(c) > -\theta + 2\xi_s$ implies $\mathrm{STR}_{P^{(k)}}(c) > -\theta$, that is, H_0^c for that epoch. If $2\xi_s \le \theta$, then $\mathrm{STR}_Q(c) > 0$ implies $\mathrm{STR}_Q(c) > -\theta + 2\xi_s$. □

Remark 1 (Data-dependent budgets). The ledger above is fixed in advance, which is what makes the union bound free of any independence assumption. A budget that depends on data (for example, α_c proportional to an estimated reuse frequency) remains valid if (i) the total is at most α almost surely and (ii) the trial sequence of cell c is independent of the information used to set α_c: then $\Pr[A_{c,k} \mid \alpha_{c,k}] \le \alpha_{c,k}$ and summing the expectations gives α. Without (ii), for instance when two cells share tasks, a data-dependent budget can break the guarantee.

Lemma 2.3 (Shift). *If f takes values in an interval of length b, then $|\mathbb{E}_Q f - \mathbb{E}_P f| \le b \cdot \mathrm{TV}(P,Q)$. Since $Z \in [-1,1]$, $|\mathrm{STR}_Q(c) - \mathrm{STR}_P(c)| \le 2\,\mathrm{TV}(P_s,Q_s) \le 2\xi_s$, and the constant 2 is attained.*

Proof. Centre f so that $|g| \le b/2$; then $|\mathbb{E}_Q g - \mathbb{E}_P g| \le (b/2)\sum_x |Q(x) - P(x)| = b\,\mathrm{TV}(P,Q)$. For attainment, take task types u with $Z = 1$ and d with $Z = -1$, $P(u) = 0.1$, $Q(u) = 0.2$: then TV $= 0.1$ and the risks differ by 0.2. □

Why the budget is per cell. Allocating one α_s per *scope* and letting every skill in the scope use the threshold $1/\alpha_s$ does not control the family-wise error: with ten skills at the null boundary, $\theta = \alpha = 0.1$ and 300 trials, the family-wise false admission rate of our e-process is 0.45, against 0.05 with the per-cell budget $\alpha/10$ (Section 2.6).

2.3 Scope and shift are necessary

Assumption 3 (Rich target class). *For every value of the features a procedure observes (fingerprints, predicates, trial outcomes on the observed repositories), there is a repository with those features whose task losses are arbitrary in $[0,1]$.*

Proposition 2.4 (No guarantee without target evidence or bounded shift). (a) *Let a procedure decide from data of observed repositories $\mathcal{R}_{\mathrm{obs}}$ only, and admit a skill S with positive probability. Under Assumption 3 there is a repository $R^\dagger \notin \mathcal{R}_{\mathrm{obs}}$ with $\mathrm{STR}(S,\{R^\dagger\}) = 1$ on which the admission probability is unchanged.* (b) *For every $\theta \in (0,1)$, $\delta > 0$ with $\theta + \delta < 1$, $\xi \in (0, 1/2]$ and every procedure, there are P, Q with $\mathrm{TV}(P,Q) = \xi$ and a cell c with $\mathrm{STR}_P(c) = -\theta - \delta$ and $\mathrm{STR}_Q(c) = \mathrm{STR}_P(c) + 2\xi$, whose admission probability does not depend on Q.*

Proof. (a) By Assumption 3, take $R^\dagger$ that has the same values of every feature the procedure reads (the data of $\mathcal{R}_{\mathrm{obs}}$ and the target's own fingerprints and predicates) and has $\ell(x,\pi) = 1$, $\ell(x,\varnothing) = 0$ for every task. The procedure's input, hence its admission probability, is unchanged, and $Z \equiv 1$ on $R^\dagger$. (b) Take task types u ($Z = 1$) and d ($Z = -1$) with $P(d) = (1 + \theta + \delta)/2 \ge \xi$, and obtain Q by moving mass ξ from d to u. The procedure only reads tasks drawn from P. □

Part (b) shows that no procedure that learns only from P can certify Q-risk better than $\mathrm{STR}_P + 2\xi$, so the shift term of Theorem 2.2 cannot be removed or reduced; this is why the guarantee is relative to a declared scope family and why coarse scopes make it weak.

2.4 Composition

Theorem 2.5 (Composition risk is not controlled by individual risks). *For $a, b \in [-1, 0)$,*

$$\sup\{\mathrm{STR}(S_1 \circ S_2) : \mathrm{STR}(S_1) = a,\ \mathrm{STR}(S_2) = b\} = 1 + \min(a,b),$$

and the supremum is attained. In particular, two skills that each meet the margin $-\theta$ can compose to excess risk $1 - \theta > 0$, and no bound on the composite risk improves on the trivial one implied by $\ell \in [0,1]$.

Proof. Upper bound, for any skills: $\mathrm{STR}(S_1 \circ S_2) \le 1 - \mathbb{E}\ell(\varnothing)$, and $\mathbb{E}\ell(\varnothing) \ge -a$ because $\mathbb{E}\ell(\pi_1) = \mathbb{E}\ell(\varnothing) + a \ge 0$; likewise $\ge -b$. Attainment, with losses realizable by Assumption 3: constant losses $\ell(\varnothing) = -\min(a,b)$, $\ell(\pi_1) = \ell(\varnothing) + a$, $\ell(\pi_2) = \ell(\varnothing) + b$, $\ell(\pi_1 \circ \pi_2) = 1$. □

Theorem 2.5 says nothing about footprints. Screening only pairs whose footprints overlap is therefore sound only under an additional assumption, which we state and test rather than presume.

Assumption 4 (Non-interaction of disjoint footprints). *For sets of skills A, B whose footprints are disjoint ($\mathcal{K}_A \cap \mathcal{K}_B = \varnothing$, with $\mathcal{K}_A$ the union over A), and every task x, $\ell(x, \pi_A \circ \pi_B) - \ell(x,\varnothing) = [\ell(x,\pi_A) - \ell(x,\varnothing)] + [\ell(x,\pi_B) - \ell(x,\varnothing)]$.*

Corollary 2.6 (What actually runs). *Under Assumption 4 and the execution rule of Section 1, outside the event of Theorem 2.2 (probability at most α), every executed set E with m footprint-connected components satisfies $\mathrm{STR}_Q(E) \le m(-\theta + 2\xi_s)$; in particular no executed set is harmful when $2\xi_s \le \theta$. Only footprint-overlapping components need a composite cell: with mean overlap degree $\bar{d}$ there are $|\mathcal{L}_s|\,\bar{d}/2$ overlapping pairs per scope.*

Proof. Components have pairwise disjoint footprints, so applying Assumption 4 $m - 1$ times and taking expectations under Q gives $\mathrm{STR}_Q(E) = \sum_{j=1}^m \mathrm{STR}_Q(E_j)$. Each E_j is an admitted cell (a single skill, or a composite by the execution rule), so outside the event of Theorem 2.2 each term is at most $-\theta + 2\xi_s$. □

Assumption 4 is strong: two edits in different functions can still interact through shared state that no footprint records. We therefore report it as an assumption and measure how often it fails on pairs with disjoint footprints, rather than presume it.

2.5 The value of pairing

Proposition 2.7 (Pairing). *Let $V_\pi = \mathrm{Var}\,\ell(x,\pi)$, $V_\varnothing = \mathrm{Var}\,\ell(x,\varnothing)$ and $C = \mathrm{Cov}(\ell(x,\pi), \ell(x,\varnothing))$ on the same task. Estimating* STR *from n paired tasks has variance $(V_\pi + V_\varnothing - 2C)/n$; from two independent arms of n tasks each it has variance $(V_\pi + V_\varnothing)/n$. Both designs cost $2n$ runs, so pairing reduces the number of tasks needed for a given precision by $R = (V_\pi + V_\varnothing)/(V_\pi + V_\varnothing - 2C)$. Under the additive model $\ell(\varnothing) = T$, $\ell(\pi) = T + S$ with independent T, S, $R = 1 + 2\sigma_T^2/\sigma_S^2$.*

Proof. Direct computation of the two variances; $V_\pi, V_\varnothing$ include any run-to-run noise, and only the shared task component enters C. Under the additive model $V_\pi + V_\varnothing = 2\sigma_T^2 + \sigma_S^2$ and $C = \sigma_T^2$. □

2.6 Numerical checks

Every statement above is checked numerically by a script in the artifact, using the e-process of our implementation: per-cell validity, the failure of a per-scope budget, the attained constant 2 in Lemma 2.3, the bound of Theorem 2.5 on random instances, the pairing ratio, the convergence of the ledger, and the failure of re-admission without a ledger (re-testing a retired cell with the full α at every epoch).

3 Related work (excerpt)

3.1 Risk control, e-processes and multiple testing

Conformal prediction [7], conformal risk control [2] and Learn-Then-Test [1] give distribution-free guarantees under exchangeability. Skill transfer violates exchangeability across repositories, so our guarantee is relative to a declared scope family (Proposition 2.4). For sequential evidence we use betting e-processes [9] and Ville's inequality [6], which make an admission valid at any data-dependent stopping time [4, 5]. Simultaneous control over the library is family-wise (FWER) and comes from a union bound over cells; it requires no independence. A false-discovery-rate variant is available through e-BH [8], at the price of a weaker, FDR-type guarantee; Benjamini–Hochberg [3] is used only for the secondary comparisons of the evaluation.

References

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